a) given MMMTT w/ h = heads $1-h$ = tails

we get: $h \cdot h \cdot h \cdot (1-h) \cdot (1-h)$

$= h^3(1-h)^2$

take log: $3 \ln h + 2 \ln(1-h)$

take $\frac{dl}{dh}$: $\frac{3}{h} - \frac{2}{1-h} = 0$

$= \frac{3}{h} = \frac{2}{1-h}$

$= 3-3h = 2h$

$= 3 = 5h$

$h = 0.6$

b) calculate $p(h \mid HHHTT)$

w/ bayes: $p(h|D) = \frac{p(D|h)\,p(h)}{p(D)}$

$p(HHHTT) = h^3(1-h)^2$

$p(h \mid HHHTT) = \frac{h^3(1-h)^2 p(h)}{p(HHHTT)} \rightarrow \int_0^1 t^3(1-t)^2 p(t)\,dt$

so $p(h \mid HHHTT) = \frac{h^3(1-h)^2 p(h)}{\int_0^1 t^3(1-t)^2 p\,t\,dt}$